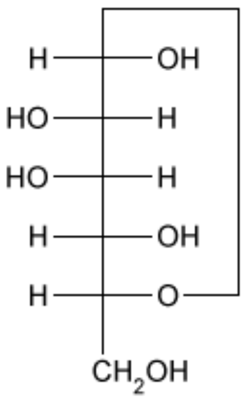
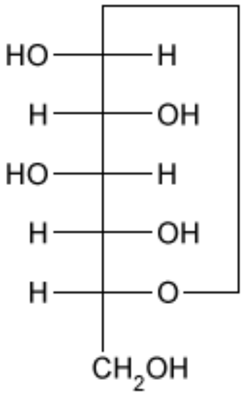
Which of the following structures represents the Fischer projection of a β-D-glucose unit found in cellulose?

A.



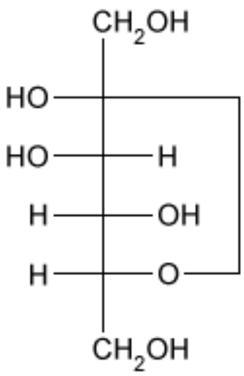
[28%]

B.



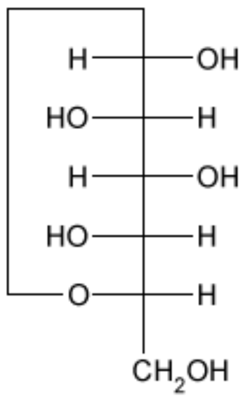
[45%]

C.



[12%]

D.



[13%]

Correct
45% answered correctly

Explanation:

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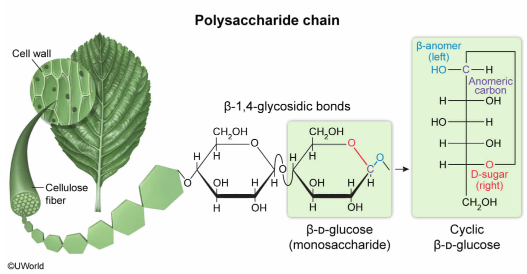

Fischer projections are two-dimensional representations of molecules that are often used to distinguish among carbohydrate **anomers, epimers, and enantiomers**.

- Carbohydrate enantiomers (L- and D-conformations) are nonsuperimposable mirror images of each other. They have the same chemical formula but with every stereocenter reversed. In a Fischer projection, D-sugars always show the second-to-last carbon (carbon 5 in glucose) as having its oxygen atom to the right. Naturally occurring sugars generally adopt the D-conformation.

- Epimers are carbohydrates that differ at **only one stereocenter**. Fischer projections of D-glucose always show one hydroxyl group to the right of carbon 2, one to the left of carbon 3, and one to the right of carbon 4. Carbohydrates that differ at exactly one of these positions are epimers of D-glucose.

- Anomers are a special case of epimers that differ from each other at the anomeric carbon. They can exist only in cyclized sugars (furanose or pyranose) because in the linear form, the anomeric carbon is a carbonyl and cannot be a stereocenter. In Fischer projections, cyclic D-sugars with a hydroxyl group to the right of the anomeric carbon are in the α-conformation, whereas those with the hydroxyl group on the left are in the β-conformation. The opposite is true for L-sugars.

Cellulose is composed of multiple β-D-glucose units in the pyranose form. Therefore, a Fischer projection of β-D-glucose must display a six-member ring with a hydroxyl group to the right of carbon 5 (D-sugar) and one to the left of the anomeric carbon (β-conformation). Of the choices available, **only the Fischer projection in Choice B has the correct stereochemistry**.

(Choice A) This structure is α-D-mannose, a glucose epimer that differs at carbon 2.

(Choice C) This structure is β-D-fructose.

(Choice D) This is β-L-glucose, the enantiomer of β-D-glucose. Because this is an L-sugar, the β anomer positions the anomeric hydroxyl on the right.

Educational objective:
Fischer projections help distinguish anomers, epimers, and enantiomers. Enantiomers are compounds with the same chemical formula but differ at every stereocenter, whereas epimers and anomers differ at only one stereocenter. Anomers exist only in cyclized sugars and differ at the anomeric carbon.

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